



Advanced Digital Image Processing Lecture #6

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Contents

- ❖ Background of Segmentation
- ❖ Color/Intensity Based Segmentation
- ❖ Region Based Segmentation
- ❖ Shape Based Segmentation
- ❖ Motion Based Segmentation



Background Of Segmentation (1 of 2)

- ❖ The objective of segmentation is to partition an image into regions based on specific criteria.
 - ❑ Color/Intensity Based Segmentation
 - ❑ Region Based Segmentation
 - ❑ Shape Based Segmentation
 - ❑ Motion Based Segmentation

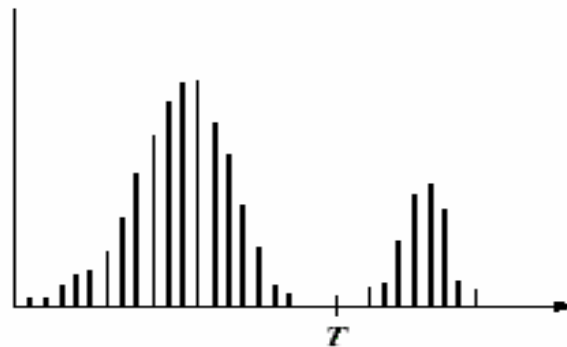


Background of Segmentation (2 of 2)

- ❖ Object point by threshold
 - ❖ Image Histogram
 - ❖ Gray Level Histogram
 - ❖ Single Threshold and Multiple Thresholds
 - ❖ Global, Local and Adaptive Threshold
- ❖ Region Based Segmentation
 - ❖ Region Growing Methods
 - ❖ Region Splitting and Merging
- ❖ Segmentation by Morphological Watersheds

Global Thresholding (1 of 2)

- ❖ Partition the image histogram by using a single global threshold, T .
- ❖ Segmentation is then accomplished by scanning the image pixel by pixel and labeling each pixel as object or background, depending on whether the gray level of that pixel is $>T$ or $\leq T$.
- ❖ Success depends on how well the histogram can be partitioned.



Gray Level Histogram

Global Thresholding (2 of 2)

Threshold Selection

- ❖ Heuristic approach, and
- ❖ Automatic threshold selection
 1. Select an initial estimate for T
 2. Segment the image using T . Two group G_1 and G_2
 3. Compute the average gray level values μ_1 and μ_2 corresponding to G_1 and G_2
 4. Compute the new threshold $T = \frac{1}{2}(u_1 + u_2)$
 5. Repeat 2-4 until the difference in T in successive iteration is smaller than predefined parameter T_0

Adaptive Thresholding

- ❖ Uneven illumination can transform a perfectly segmentable histogram into a histogram that cannot be partitioned effectively by a single global threshold.
- ❖ To handle uneven illumination,
 - ❑ Divide the original image into sub-images
 - ❑ Use different threshold to segment each sub-image.Continue this process so that the illumination of each sub-image is approximately uniform.
- ❑ Threshold for each pixel depends on the position of the pixel in the sub-image.

Region Based Segmentation (Basic Formulation)

- ❖ Let R represent the entire image region and $R_1, R_2, \dots, R_n$, are subregions

(a)
$$\bigcup_{i=1}^n R_i = R$$

(b) R_i is a connected region, $i = 1, 2, \dots, n$

(c) $R_i \cap R_j = \phi$ for all i and j , $i \neq j$ // ϕ Null set (regions are disjointed)

(d) $P(R_i) = \text{True}$, for $i = 1, 2, \dots, n$ // $P(R_i)$ logical predicate defined over the points in set R_i , properties satisfied by the pixels in a segmented region.

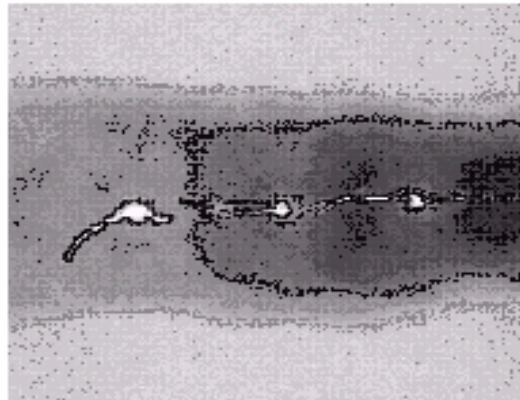
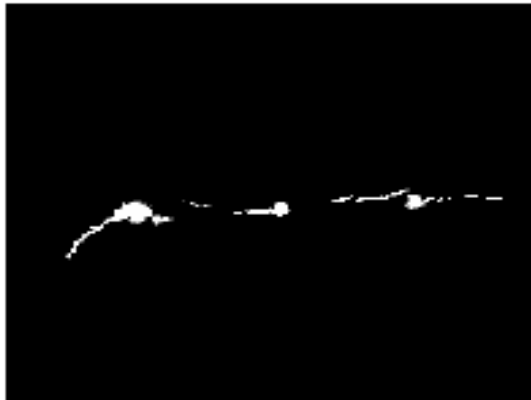
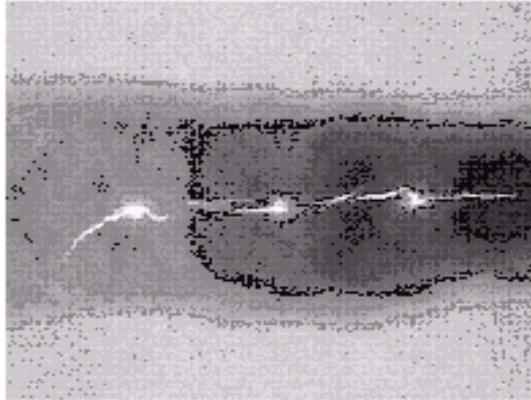
(e) $P(R_i \cup R_j) = \text{False}$ for $i \neq j$ // Two regions are different in the sense of predicate P . // one condition is not valid in other regions

Region Growing (1 of 2)

- ❖ Group pixels to sub-regions into a larger regions based on predefined criteria.
 - ❑ Start with a set of “seed” points and from these grow regions by appending to each seed those neighboring pixels that have properties similar to the seed (specific range of gray level or color).
- ❖ Limitations:
 - ❑ Formulation of Stopping rule
 - ❑ When no more pixels satisfy the criteria growing region should be stopped.
 - ❑ do not consider the history of region growth – size and shape

Region Growing (2 of 2)

a, b
c, d

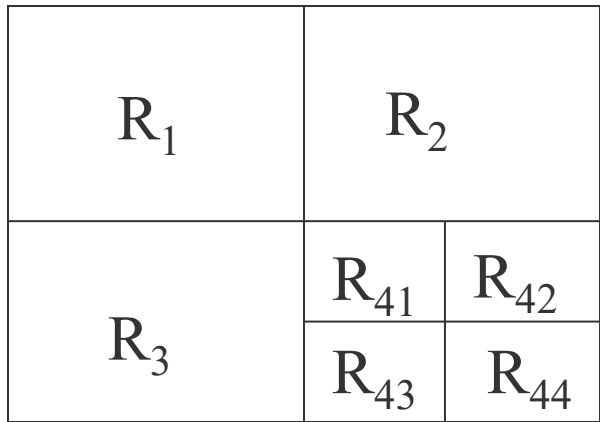


- Seed points gray value=255
- Difference between seed point with other points gray level is less than 65
- Find connected region that satisfies same condition

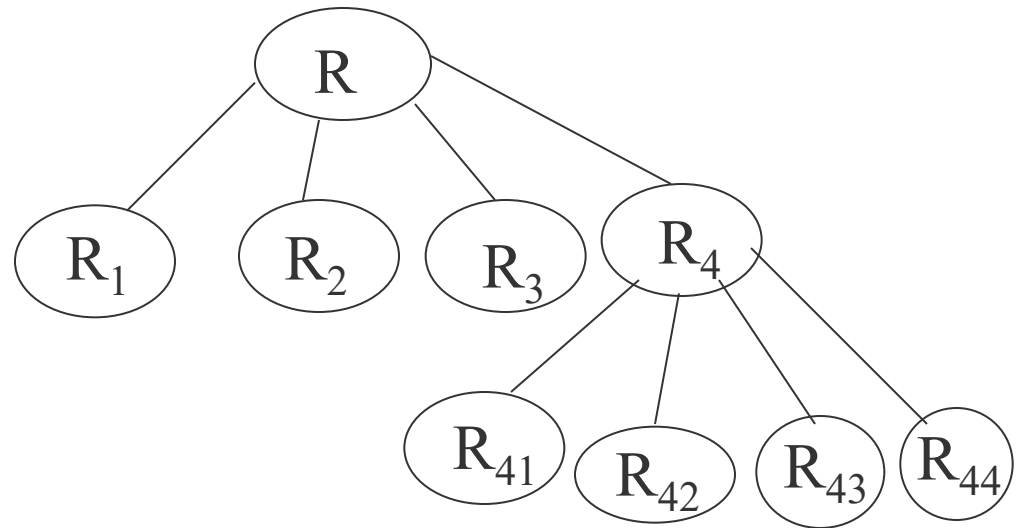
a) Defective welds, b) Seed points c) Result of region growing, d) Boundaries of segmented defective welds

Region Splitting and Merging (1 of 2)

- ✦ Partition the image into numbers of quadrant regions



Partition image



Quadtree



Region Splitting and Merging (2 of 2)

- ❖ Split into four disjoint quadrants any region R_i for which $P(R_i) = \text{False}$.
- ❖ Merge any adjacent regions R_j and R_k for which $P(R_j \cup R_k) = \text{True}$
- ❖ Stop when no further merging or splitting is possible



Morphological Watershed (1 of 4)

The concept of watersheds is based on visualizing an image in three dimension: two coordinates versus gray levels. To explain this we consider 3 points:

- (a) Points belonging to a regional minimum;
- (b) Points at which a drop of water, if placed at the location of any of those points, would fall with certainty to a single minimum; (such points called catchment basin or watershed of that minimum)
- (c) Points at which water would be equally likely to fall to more than one such minimum (Watershed lines)



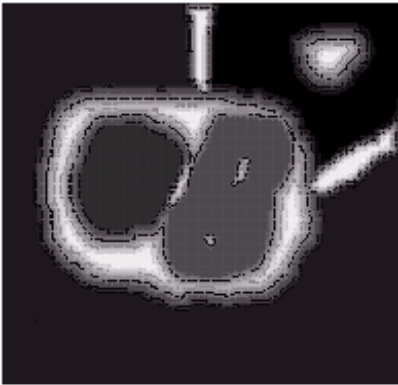
Morphological Watershed (2 of 4)

Find the watershed lines:

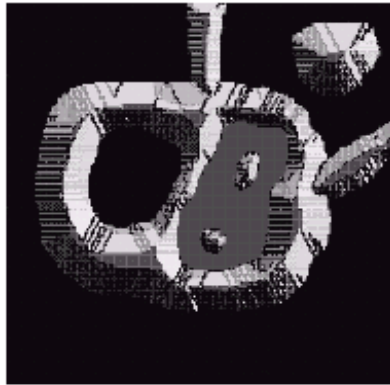
- (a) Punch a hole in each regional minimum
- (b) Flood the entire topography from below by letting water rise through the holes at a uniform rate.
- (c) When the rising water in distinct catchment basin is about to merge, build dams to prevent catchment basin from merging
- (d) Once fully flooded, only the top of the dams are remained.
- (e) These dams (closed) boundaries corresponding to the divide lines of the watersheds.

Morphological Watershed (3 of 4)

A



B

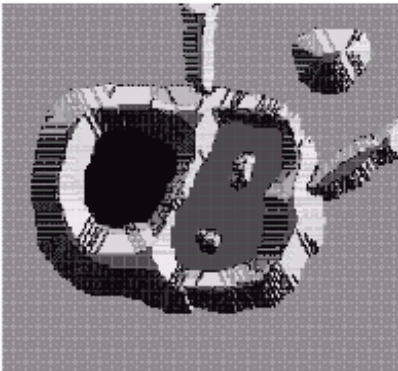


(A) = Original Image

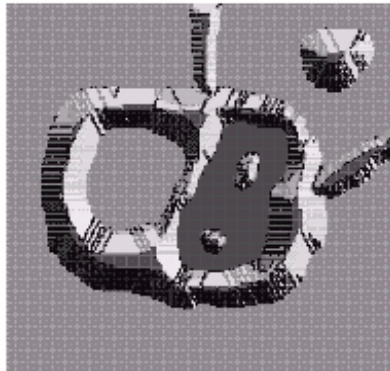
(B) = Topographic view

(C), (D) = Two stages of
flooding

C

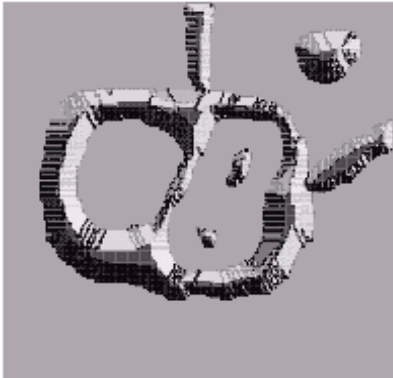


D

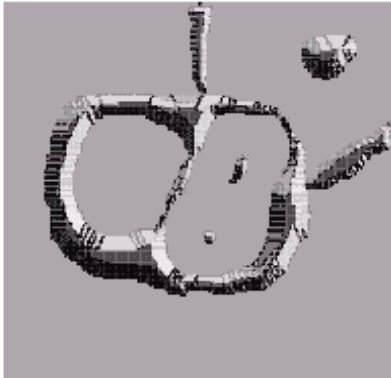


Morphological Watershed (4 of 4)

E

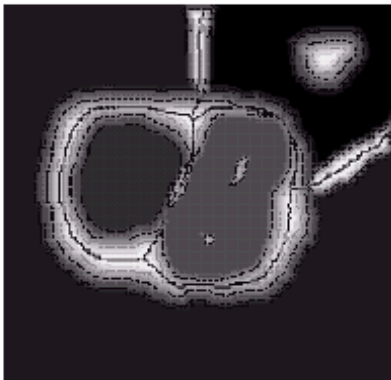
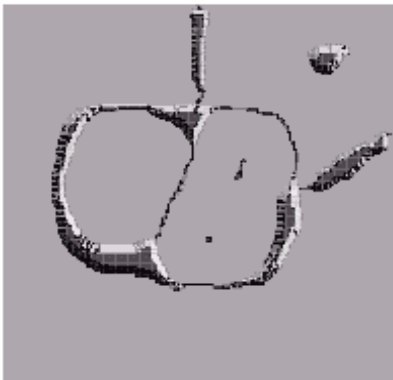


F



E. Result of Further Flooding

F. Beginning of Merging of Water from two catchment basin



G

H

G. Longer Dams

H. Watershed (segmentation lines)



Motion Based Segmentation

- ❖ Background subtraction from each image frame: when camera and background is fixed

$$M_i = X_t - B \text{ (B=background image)}$$

- ❖ Subtraction two successive image frames: if camera or background or both is moving.

$$M_i = X_t - X_{t+1}$$



List of Reference Books

- ❖ Rafael C. Gonzalez, “Digital Image Processing”.
- ❖ Anil K. Jain, “Fundamental of Digital Image Processing”.
- ❖ Dana H. Ballard, “Computer Vision”
- ❖ David H. Hubel, “Eye, Brain, and Vision”
- ❖ Dwayne Phillips, “Image Processing in C”



End of Today's Lecture

Questions and Suggestions?

Thanks to all!